

TRIVIAL UTF-8 MANUAL

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[in package TRIVIAL-UTF-8]

1 Introduction

Trivial UTF-8 is a small library for doing UTF-8-based in- and output on a Lisp implementation that already supports Unicode - meaning `char-code` and `code-char` deal with Unicode character codes.

The rationale for the existence of this library is that while Unicode-enabled implementations usually do provide some kind of interface to dealing with character encodings, these are typically not terribly flexible or uniform.

The `Babel` library solves a similar problem while understanding more encodings. Trivial UTF-8 was written before Babel existed, but for new projects you might be better off going with Babel. The one plus that Trivial UTF-8 has is that it doesn't depend on any other libraries.

2 Links and Systems

The official repository is <https://gitlab.common-lisp.net/trivial-utf-8/trivial-utf-8>, and this document is available in various formats on <https://fixnum.com> for the latest version.

- `[system] "trivial-utf-8"`
 - *Description:* A small library for doing UTF-8-based input and output.
 - *Licence:* ZLIB
 - *Author:* Marijn Haverbeke marijnh@gmail.com

- *Maintainer:* Gábor Melis mega@retes.hu
- *Homepage:* <https://common-lisp.net/project/trivial-utf-8/>
- *Bug tracker:* <https://gitlab.common-lisp.net/trivial-utf-8/trivial-utf-8/-/issues>
- *Source control:* [GIT](#)
- *Depends on:* `mgl-pax-bootstrap`

3 Reference

- **[function]** `utf-8-byte-length` *string*
Calculate the amount of bytes needed to encode *string*.
- **[function]** `string-to-utf-8-bytes` *string &key null-terminate*
Convert *string* into an array of unsigned bytes containing its UTF-8 representation. If *null-terminate*, add an extra 0 byte at the end.
- **[function]** `utf-8-group-size` *byte*
Determine the amount of bytes that are part of the character whose encoding starts with *byte*. May signal `utf-8-decoding-error`.
- **[function]** `utf-8-bytes-to-string` *bytes &key (start 0) (end (length bytes))*
Convert the *start*, *end* subsequence of the array of bytes containing UTF-8 encoded characters to a `string`. The element type of *bytes* may be anything as long as it can be `coerced` into an `(unsigned-bytes 8)` array. May signal `utf-8-decoding-error`.
- **[function]** `read-utf-8-string` *input &key null-terminated stop-at-eof (char-length -1) (byte-length -1)*
Read UTF-8 encoded data from *input*, a byte stream, and construct a string with the characters found. When *null-terminated* is given, stop reading at a null character. If *stop-at-eof*, then stop at `end-of-file` without raising an error. The *char-length* and *byte-length* parameters can be used to specify the max amount of characters or bytes to read, where -1 means no limit. May signal `utf-8-decoding-error`.
- **[function]** `write-utf-8-bytes` *string byte-stream &key null-terminate*
Write *string* to *byte-stream*, encoding it as UTF-8. If *null-terminate*, write an extra 0 byte at the end.
- **[condition]** `utf-8-decoding-error` *simple-error*

4 Indices

Referrer definition type abbreviations:

- *f*: for definitions in the function namespace (macros, compiler macros and also methods)

- *t*: DEFTYPEs, classes, conditions, structs
- *d*: documentation sections and glossary terms
- *l*: definitions of definition types
- *s*: ASDF systems
- *p*: packages
- *n*: named readtables
- *v*: special variables and constants
- *r*: restarts
- *?*: other

4.1 Function and Macro Index

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[write-utf-8-bytes](#) 2 (*fn*)

4.2 Type Index

[utf-8-decoding-error](#) 2 (*condition*) $\leftrightarrow$ *f*: [read-utf-8-string](#) 2, [utf-8-bytes-to-string](#) 2,
[utf-8-group-size](#) 2

4.3 Misc Index

[trivial-utf-8](#) 1 (*asdf:system*)